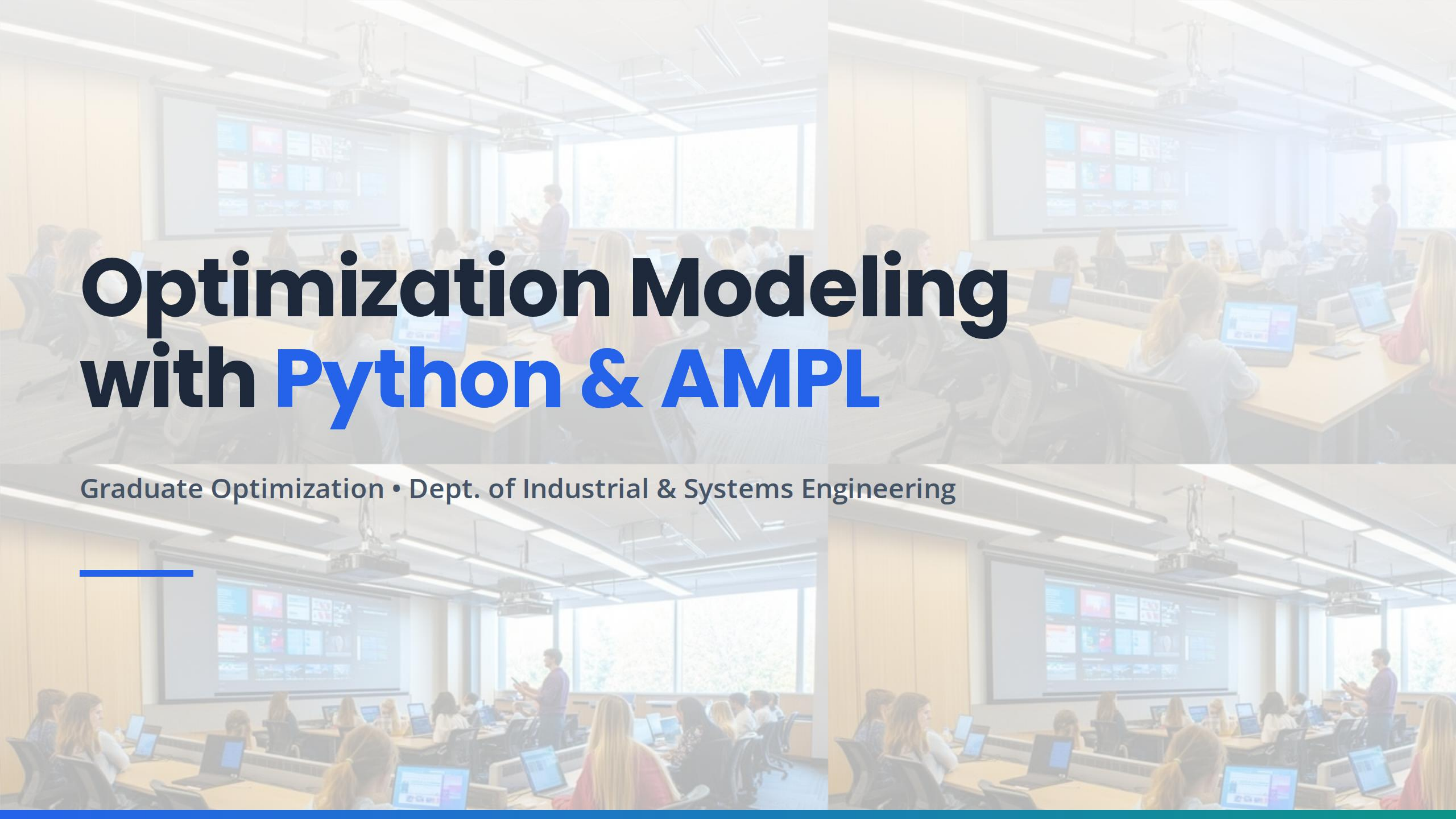


The background of the slide is a collage of four images showing a modern lecture hall. In each image, students are seated at long wooden tables, working on laptops. At the front of the room, a large screen displays a grid of colorful icons. A lecturer is visible at the front, interacting with the screen. The room has large windows and modern lighting.

Optimization Modeling with **Python & AMPL**

Graduate Optimization • Dept. of Industrial & Systems Engineering

The Optimization Ecosystem



Spreadsheets

Simple & accessible. Good for prototypes. **Limit:** Poor scalability, hard to debug, weak solvers.



Python Libraries

PuLP , **SciPy** . Good for automation. **Benefit:** Integrates data cleaning and solving.



Commercial Solvers

Gurobi & CPLEX. The industry standard. Capable of solving millions of constraints efficiently.

Software Access for NIU Students

Academic Licenses

- ✓ **Gurobi:** Free "Named User Academic License" at *gurobi.com* with your **@niu.edu** email.
- ✓ **IBM CPLEX:** Available via the IBM Academic Initiative.

Cloud Access

- ✓ **AMPL:** I have provided the course access code for the full version.
- ✓ **Google Colab:** AMPL Community Edition is built-in and free. No installation required.

Why use a Modeling Language (AMPL)?

Separation of Model & Data:

-  **The Model:** Abstract logic. Written once.
-  **The Data:** Specific numbers. Updated frequently.
-  **The Solver:** Swap engines (Gurobi, Highs, CBC) with one line of code.

```
# AMPL creates an abstraction
var Ship {Origins, Dests} >= 0;

maximize Profit:
    sum {i in Origins}
        Profit[i] * Ship[i,j];

# Focus on logic, not matrix gen
```

Case Study: Land Allocation

Forest Service Management

Goal: Allocate 1,000 acres of public land between two uses to maximize Net Present Value (NPV).

- 🌲 **Use 1: Timber (X_1)**
High return (\$500/acre), High cost.
- 🌿 **Use 2: Preservation (X_2)**
Social value (\$200/acre), Low cost.
- 📊 **Constraints:** Total Land, Budget (\$50k), Min Preservation (200 acres).



Mathematical Formulation

Decision Variables

X_1 : Acres allocated to Timber

X_2 : Acres allocated to Preservation

Objective Function

Maximize $Z = 500X_1 + 200X_2$

Subject to:

$$1X_1 + 1X_2 \leq 1000 \text{ (Land)}$$

$$100X_1 + 20X_2 \leq 50000 \text{ (Budget)}$$

$$X_2 \geq 200 \text{ (Policy)}$$

$$X_1, X_2 \geq 0$$

Setup in Google Colab

Initialization

To run AMPL in the cloud (Colab), we use the `ampl_notebook` helper. This ensures everything is configured correctly for the cloud environment.

We then install our chosen solvers (e.g., HiGHS).

```
# Install AMPL Python API
!pip install amplpy -q
from amplpy import AMPL, ampl_notebook

# Initialize AMPL for Colab
ampl = ampl_notebook(
    modules=["highs"],
    license_uuid="your-license-uuid"
)
```

The AMPL Pipeline: 2 Key Choices



1. Defining the Model

How do we construct variables and constraints?

Option A: Inline (`ampl.eval`)

Option B: File (`.mod`)



2. Loading the Data

How do we assign values to parameters?

Option A: Python (Pandas/Dict)

Option B: File (`.dat`)



3. Solving

This step remains the same regardless of previous choices.

`ampl.solve()`

Modeling Method A: Inline (`ampl.eval`)

The "Notebook" Approach

Pass the model structure as a string.

- ✓ Great for teaching.
- ✗ Harder to debug.

```
ampl.eval('''  
    set USES;  
    param return_per_acre {USES};  
    param budget_limit;  
  
    var Acres {u in USES} >= 0;  
  
    maximize Total_NPV:  
        sum {u in USES}  
            return_per_acre[u] * Acres[u];  
''')
```

Modeling Method B: Separate File (.mod)

The "Production" Approach

Save logic in `resource.mod`.

- ✓ Reusable.
- ✓ Syntax highlighting.

```
# In Python:  
from amplpy import AMPL  
ampl = AMPL()  
  
# Load the model file  
ampl.read('resource.mod')
```

Data Method A: Direct Injection (Python)

Using Python Data

Load Python/Pandas data into AMPL.

```
# Python Data
uses = ['Timber', 'Preservation']
ret = {'Timber': 500, 'Preservation': 200}

# Inject into AMPL
ampl.getSet('USES').setValues(uses)
ampl.param['return_per_acre'] = ret
ampl.param['budget_limit'] = 50000
```

Data Method B: External File (.dat)

Using AMPL Data Syntax

Data stored in `resource.dat` using `:=` syntax.

```
# resource.dat  
param return_per_acre :=  
    Timber 500  
    Preservation 200;  
  
param budget_limit := 50000;
```

```
# Python  
ampl.read_data('resource.dat')
```

Step 4: Solve & Analyze

Solving

```
ampl.option['solver'] = 'highs'  
ampl.solve()
```

Get Results

```
print(ampl.obj['Total_NPV'].value())  
df = ampl.var['Acres'].get_values().to_pandas()
```

Interpreting: Shadow Prices

What is a Shadow Price?

Rate of improvement in Objective per unit increase in RHS.

Example: If `Budget.dual` = 3.5, then +\$1 Budget = +\$3.50 NPV.

Code to Access Duals

```
# Access the .dual suffix
shadow_price = ampl.con['Budget_Limit'].dual()

print(f"Marginal Value: {shadow_price}")
```

Troubleshooting: Solver Status

Optimal

Best solution found. Proceed.

Infeasible

No solution satisfies all constraints. (e.g., $\text{Min} > \text{Max}$).

Unbounded

Objective goes to infinity. (Missing constraint?).

Resources & Next Steps

-  **Colab Notebook:**
Available on Blackboard.
-  **AMPL Book:**
Chapter 2 covers `.dat` files.
-  **Assignment:**
Re-implement "Ohio National Bank" model.



Q&A

Questions regarding Colab setup, Solver installation, or AMPL?

Image Sources



<https://cdn.sanity.io/images/hvd5n54p/production/96118674f70123800c3adeea5fd692bf914f4e29-1920x1494.jpg?auto=format>

Source: www.planet.com



https://images.stockcake.com/public/2/9/6/29614363-8879-4c79-a589-cf51a9fe0c74_large/interactive-learning-session-stockcake.jpg

Source: stockcake.com